Boomilever — Division C Nationals Build Guide

Target: Tier 1, bonus geometry, 15 kg held, ~8 g structure, structural efficiency ≈ 2400–2600.

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0. Read this before anything else

This guide is written against the **2026 Division C rules**, the most recent published set. The **2027 rules release September 8, 2026**, and Science Olympiad's own July bootcamp closed with "Next Year will be Boomilevers with slight changes to rules."

Do not cut wood until you have read the 2027 rules. The engineering below is rule-agnostic; the *numbers* are not. On September 8, check these seven things and update the dimensions in § 3 accordingly:

#	Check	2026 Div C value
1	Length range, wall to Load Block center	40–45 cm
2	Contact Depth limit (all touch points above this line)	< 15 cm
3	**Bonus** Contact Depth limit	< 10 cm
4	Contact Width requirement	**> 8 cm**
5	Load Block bottom constraint prior to loading	above the CD line
6	Maximum load / bonus value	15 kg / +5,000 g
7	Scoring formula and tier definitions	(Load + Bonus) ÷ Mass

Note item 4 is a *minimum*, and it is inverted between divisions — Division B requires contact width **under** 8 cm, Division C **over** 8 cm. Teams that reuse a Div B design get an instant Tier 2. Verify this hasn't flipped again.

Also confirm: the Loading Block is 5×5×2 cm with an 8 mm center hole, eye bolt ¼", 15.2 kg maximum applied, 6-minute test window, ANSI Z87+ eye protection mandatory.

One thing I could not verify from the sources I have is the exact required interface between your structure's distal end and the Loading Block — whether the block sits on top of a platform, is captured underneath, or hangs from a hole. Section 7 of the rules and the official diagram govern. Build a mock-up of the block and eye bolt from scrap and physically test the fit before committing. This is the single most common source of a Tier 3 "could not be tested."

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1. Why the bonus is worth taking (and when it isn't)

This decision drives the entire design, so make it first, with arithmetic rather than instinct.

Geometry and forces

Set the wall as the plane $x = 0$, with y measured **downward** from the mounting bolt.

- Mounting bolt (Attach Point): (0, 0)
- Wall touch points: (0, CD)
- Load Block center: (L, CD − 1), since the block is 2 cm tall and its bottom sits on the CD line

The structure is a simple two-force triangle. With load W hanging at the end, θ the angle between the tension member and the compression member:

```
θ  = arctan( (CD − 1) / L )
Compression  ≈ W / tan θ
Tension      ≈ W / sin θ
```

Running both cases at L = 40 cm, W = 15 kg:

	Non-bonus (CD = 15)	Bonus (CD = 10)
Effective rise	14 cm	9 cm
θ	19.29°	12.68°
Compression	42.9 kg (421 N)	66.6 kg (653 N)
Tension	45.4 kg (445 N)	68.3 kg (670 N)

Bonus geometry raises member forces by ~55 %. That is the price. The prize is +5,000 g on a 15,000 g load — a 33 % score increase.

The break-even

Score with bonus beats score without whenever

```
(15000 + 5000) / M_bonus  >  15000 / M_plain
M_bonus < 1.333 × M_plain
```

So the bonus pays if the heavier structure comes in under 1.333× the mass of the lighter one. Does it?

For **buckling-critical** compression members, Euler gives $P_{cr} \propto EI/L^2$, and for a square section $I = A^2/12$, so $P \propto A^2$. Therefore $A \propto \sqrt{P}$ and **mass** $\propto \sqrt{P}$. A 1.55× force increase costs only $\sqrt{1.55} = $ **1.25× mass**.

For **crushing-critical** members and for tension members, stress is F/A, so mass scales linearly: **1.55×**.

Real structures are mixed. Compression dominates the mass budget and is largely buckling-driven once you brace properly, so the blended figure lands around **1.28–1.32×** — just inside the 1.333 threshold.

The honest conclusion

Take the bonus, but only if you can reliably hold the full 15 kg. The margin is thin, and it collapses entirely on a partial hold. A bonus-geometry structure that breaks at 14.5 kg scores 14500/M_bonus — roughly 20 % *worse* than a non-bonus build that held 15 kg. There is no partial credit on the bonus.

Decision rule: attempt the bonus at Nationals only if, in your last five test builds of the final design, **at least four held 15 kg without failure**. If you're at 3-of-5 or worse, build the non-bonus geometry and win on mass instead. A reliable 15 kg at 7 g (SE 2143) beats a coin-flip 20 kg at 8 g.

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2. Materials

Balsa is not one material

Balsa density ranges from about 0.08 to 0.25 g/cm³ — a threefold spread — and its properties scale nonlinearly with density:

- Young's modulus **$E \propto \rho^{1.5}$** approximately (2.0 GPa at 0.10 g/cm^3, ~4.5 GPa at 0.18)
- Compressive strength along grain **$\sigma_c \propto \rho^{1.3}$**, roughly 8 MPa at 0.10, 16 MPa at 0.18
- Tensile strength along grain **σ_t** roughly 2× σ_c

The figures of merit differ by loading mode, and this is why you do not use one density everywhere:

Member	Failure mode	Figure of merit	Wanted density
Compression, well braced	Euler buckling	$E^{0.5} / \rho$	**Light**, 0.09–0.12
Compression, short bays	Crushing	σ_c / ρ	**Medium**, 0.14–0.18
Tension	Tensile / glue joint	σ_t / ρ	**Medium-dense**, 0.15–0.20
Webs, bracing	Stiffness only	E / ρ	**Lightest available**

What to buy

- **Balsa:** 3/16" × 3/16" and 1/8" × 1/8" sticks, 1/16" × 3/8" strips, 1/16" sheet. Buy from a supplier that sorts by density — Specialized Balsa, Balsa USA, or a competition-grade seller. Buy **three to four times** what you think you need; you will reject most of it.
- **Hardwood:** a small piece of 1/8" birch ply or basswood for the attach block.
- **Glue:** thin cyanoacrylate (CA) for the primary joints, medium CA for gap-filling at the attach block. Accelerator. Avoid epoxy — it is far heavier for the strength gained at these scales.
- **Tools:** razor saw or fresh #11 blades, steel rule, sanding block, jeweler's scale (0.01 g resolution), digital calipers.

The screening protocol — this is where the event is won

Most teams skip this. Do not.

1. **Weigh and measure every stick.** Compute $\rho = m / (L \times w \times h)$. Label each stick with its density in pencil.
2. **Sort into density bins** of 0.02 g/cm^3 and store in labeled envelopes.
3. **Reject on grain.** Discard anything with visible grain runout (grain lines exiting the side of the stick within its length), knots, discoloration, or soft spots. Runout in a tension

member is a guaranteed failure at a fraction of rated load.

4. **Flex test.** Support a stick at both ends, press the center lightly, release. Stiff sticks that spring back crisply are what you want. Anything that feels dead or takes a set goes in the scrap bin.
5. **Pair by density.** Left and right compression legs must come from sticks within **0.01 g/cm³** of each other. Mismatched legs load unevenly, and the stiffer one takes more than half the force and buckles early. This one step is worth more than any geometry tweak.

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3. The design

Configuration: horizontal compression A-frame + twin diagonal tension members.

In plan view (looking down), two compression legs splay from the load end back to two wall touch points, forming a triangle. In side view, two tension members run from the mounting bolt down and out to the same load end. The compression legs are the heavy structure; the tension members are thin.

Principal dimensions (2026 Div C, bonus geometry)

Parameter	Value	Reason
Length, wall to Load Block center	**40.5 cm**	Minimum legal is 40; the 5 mm is insurance against measurement dispute. Every extra cm raises forces.
Contact Depth	**9.3 cm**	Bonus needs < 10; 7 mm of margin for build tolerance and wall irregularity.
Contact Width (touch point spread)	**9.0 cm**	Must exceed 8 cm; 1 cm margin.
Load Block bottom	on the 9.3 cm plane	Must be above the bonus line.
Compression leg length	41.5 cm each	Hypotenuse including the plan-view splay.
Tension member length	41.5 cm each	$\sqrt{(40.5^2 + 8.3^2)}$

Truss depth (vertical)	2.5 cm at wall, tapering to 1.5 cm at load end	Bending is highest at the wall.

Member sizing, with the calculations

Compression legs. Total compression 653 N, split between two legs = **327 N each**. Apply a 1.3 design factor → 425 N.

Buckling check. With bracing every 5 cm (bay length $L_b = 0.05$ m), pinned-pinned:

```
I_required = P·L_b² / (π²·E)
           = 425 × 0.0025 / (9.87 × 3.0×10⁹)
           = 3.6×10⁻¹¹ m⁴ = 36 mm⁴
For a square section, I = a⁴/12  →  a = 4.6 mm
```

Crushing check. At a = 4.76 mm (3/16"), A = 22.7 mm^2, stress = 327/22.7 = **14.4 MPa**.

That exceeds the ~9 MPa capacity of 0.10 g/cm^3 balsa. **Crushing governs, not buckling** — which is the single most important sizing insight in this event, and the reason naive Euler-only designs fail early.

Resolution: use **0.16 g/cm³ balsa** ($\sigma_c \approx 15$ MPa) at 3/16" square. Utilization 14.4/15 = 96 % — tight, so either step to 0.18 g/cm^3 or laminate two 3/32" × 3/16" strips with the glue line vertical, which also arrests grain-runout failures.

- Mass per leg: 41.5 cm × 22.7 mm^2 × 0.16 g/cm^3 = **1.51 g**
- Two legs: **3.02 g**

Tension members. Total 670 N, split two ways = **335 N each**, ×1.3 → 436 N.

Balsa at 0.16 g/cm^3 has $\sigma_t \approx 28$ MPa. A = 436/28 = **15.6 mm²**. Use **1/16" × 3/8"** (1.6 × 9.5 = 15.2 mm^2), oriented with the wide face vertical so it resists the small out-of-plane bending near the attach block.

- Mass per member: 41.5 cm × 15.2 mm^2 × 0.16 = **1.01 g**
- Two members: **2.02 g**

Tension members almost never fail in the wood. They fail **at the glue joint**. See § 5.

Bracing. 1/16" × 1/8" balsa at the lightest density you have, 0.09–0.11 g/cm^3. Bays every 5 cm along each leg, plus lateral cross-ties between the two legs at every third bay, plus diagonal shear bracing in the plan of the A-frame. Approximately 55 cm of total stick.

- Mass: ≈ **0.9 g**

Attach block. 1 × 1 × 2 cm hardwood is the template suggestion but is heavier than necessary. Use 1/8" birch ply, 2 cm × 2.5 cm, with the bolt hole drilled and both faces relieved to reduce mass. ≈ **0.5 g**

Load end fitting. A small platform of 1/16" sheet, cross-grained, to spread the block reaction into both legs. ≈ **0.3 g**

Mass budget

Item	Mass
2 compression legs	3.02 g
Bracing and webs	0.90 g
2 tension members	2.02 g
Attach block	0.50 g
Load end fitting	0.30 g
Glue (see § 5 — budget it, it is real)	1.00 g
Total	**7.74 g**

Predicted score: (15,000 + 5,000) / 7.74 = **2,584**

If glue creeps to 2 g — very easy — you land at 8.74 g and SE 2,289. **Glue discipline alone is worth ~300 points.**

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4. Build the jig first

Do not attempt freehand assembly. The jig is 80 % of the accuracy.

Materials: a flat 60 × 30 cm piece of MDF or particle board, wax paper, pins, small squares.

1. **Draw the plan view full-size** directly on the board: centerline, wall line, the two touch points at ±4.5 cm, the load end, both compression legs.
2. **Build a simulated test wall.** A vertical piece of ply screwed square to the base, with a bolt through it at the correct height and horizontal lines drawn at 9.3, 10, and 15 cm below the bolt, plus vertical lines at ±4 cm. Verify square with a machinist's square, not by eye.
3. **Cover the plan drawing with wax paper.** CA does not bond to it.
4. **Cut spacer blocks** to hold the truss at 2.5 cm and 1.5 cm heights at the correct stations.

Verify the jig against a ruler twice before you build on it. A 2 mm jig error becomes a Tier 2 disqualification at check-in.

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5. Assembly

Glue technique — read this twice

Glue is typically **10–15 % of finished mass** and contributes almost nothing structurally in excess. Every drop past what wets the joint is pure score loss.

- Use **thin CA**, applied with a fine applicator tip or a pin — never straight from the bottle.
- **Wick, don't puddle.** Bring the joint together dry, then touch a trace of CA to the seam and let capillary action pull it in.
- Balsa end grain drinks glue. **Seal end-grain joints** with a trace of CA, let it cure, then glue the joint. This halves consumption.
- Use accelerator sparingly — it embrittles the bond. Let joints cure naturally where time allows.
- **Weigh your structure at every stage** against the budget in § 3. If you are over at step 4, you will be over at the end.

Sequence

Step 1 — Compression legs. Cut both legs to 41.5 cm from density-matched stock. If laminating, glue the two halves under light even pressure on a flat surface, with wax paper beneath. Cure fully before proceeding.

Step 2 — Leg truss webs. Pin one leg to the jig. Add vertical web members every 5 cm, cut to the tapering height (2.5 cm at the wall end to 1.5 cm at the load end). Then add the diagonal web in each bay. Diagonals should run so they are in **tension under load** — for a cantilever loaded downward at the free end, diagonals slope **down toward the wall**. A diagonal in compression buckles at a fraction of its tensile capacity.

Repeat for the second leg. Build both on the same jig so they are identical.

Step 3 — Join the A-frame in plan. Set both legs on the plan drawing, converging at the load end and splaying to the ±4.5 cm touch points. Glue the load-end junction. Add lateral cross-ties between the legs every third bay (≈ 15 cm) and diagonal plan bracing between them. This is what stops the two legs from splaying or scissoring under load.

Step 4 — Touch pads. At each wall end, add a small cross-grained 1/16" pad, roughly 8 × 8 mm, square to the wall plane. These distribute the contact force and stop the leg end from crushing into the wall. **Confirm on the jig wall that they sit above the 9.3 cm line with margin.**

Step 5 — Attach block. Drill the bolt hole slightly oversize, chamfer both sides. Mount to the simulated wall bolt.

Step 6 — Tension members. This is the highest-risk joint in the structure. With the compression A-frame mounted on the jig wall and the load end supported at the correct height, cut each tension member to fit from the attach block to the load end.

Critical joint detail: do not butt-joint the tension member to the attach block. Instead:

- **Lap the tension member along the block** for at least 15 mm of face-grain contact, and
- **Wrap the joint** with a light thread or fine glass tissue set in CA, or
- Better: run a **single continuous tension member** from the load end, up around the attach block, and back down to the load end, so the bolt load is carried in a loop rather than through a glue line in shear.

A butt joint here will fail near 40 % of rated load and it will look like a mystery failure on video. It isn't — it's the glue line.

Step 7 — Load end fitting. Add the cross-grained platform. **Test-fit the actual Loading Block and eye bolt now.** Confirm the block's bottom sits above the CD line and that the assembly can be mounted within the 6-minute window without fiddling.

Step 8 — Final inspection. Weigh. Sight down every member for straightness. Check symmetry with calipers, left versus right, at five stations. Verify every dimension in §3 against the jig wall.

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6. Test, diagnose, iterate

Build a compliant test stand. You cannot optimize what you cannot measure. The stand needs the wall with correct lines, a mounting bolt, and a way to apply load progressively — a sand hopper with a valve, or careful manual pouring.

Budget five to eight builds across the season. Nobody's first structure is competitive.

The slow-motion protocol

This is the highest-value diagnostic technique in build events and costs nothing.

1. Film every load test in **slow motion** on a phone, framing the whole structure.
2. After failure, trim the clip to one second either side of the break.
3. Step **frame by frame** to the first visible failure. Everything after is collateral damage — a tension member snapping after the compression leg buckled tells you nothing.
4. Screen-grab the initiating failure. Log it.

Failure log

Keep a table: build number, mass, load held, SE, initiating failure location, failure mode, and the one change made for the next build. **Change one variable at a time.**

Reading the failure

What you see	Diagnosis	Fix
Compression leg bows sideways between bays	Euler buckling, bay too long	Shorten bay spacing, or add lateral cross-ties
Leg crushes/mushrooms at the wall pad	Crushing, density too low	Higher density stock, or larger section at the wall end
Legs scissor or splay apart	Insufficient plan bracing	Add diagonal bracing between legs
Tension member pulls off the attach block	Glue joint in shear	Lap and wrap the joint, or use a continuous loop
Tension member snaps mid-span	Grain runout, or undersized	Rescreen wood; increase section

Whole structure twists	Asymmetry — mismatched legs	Density-pair the legs; check jig square
Failure well below prediction, clean break	Grain runout in a primary member	Tighten screening

If failures are **at the wall pads**, you are crushing-limited and should go denser. If they are **mid-bay**, you are buckling-limited and should brace tighter. These call for opposite fixes, which is exactly why the slow-motion diagnosis matters.

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7. Competition day

Transport. Build a rigid box, foam-lined, that supports the structure at the attach block and load end only. Most losses happen in the car. Carry it yourself; never check it or let it ride loose.

Bring: the structure, a backup structure if the rules allow one entry per team (they permit only one *entered*, but a spare in the box protects against transport damage before check-in), ANSI Z87+ eye protection for every competitor, and your load estimate.

Check-in.

- Weigh the structure and record the mass.
- Submit your **estimated load supported** — this is a tie-breaker in the 2026 rules, so estimate honestly from your test history rather than optimistically.
- Declare whether you are attempting the bonus.
- Let the supervisor complete compliance checks without argument.

The 6-minute window. Practice this until it is automatic:

1. Mount the structure on the bolt. Seat it. Confirm touch points are above the line.
2. Mount the Loading Block assembly.
3. Attach the empty bucket.
4. Supervisor verifies. **Do not rush this step** — a compliance failure here is a Tier 2 and costs more than any loading speed gain.
5. Load steadily. Steady beats fast: sudden pours create dynamic loads well above the static weight.

Do not touch the bucket with your hands. Stabilization sticks are permitted to stop sway but must not lift or push. Many top teams skip them — a stick that momentarily bears load can void the run.

After the test, if time permits, ask the supervisor for their read on the failure. They see hundreds of structures and will tell you things no amount of solo testing will.

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8. Where the points actually are

Ranked by score impact per hour invested:

1. **Wood screening and density pairing** — routinely worth 20–30 % SE, and costs only patience.
2. **Glue discipline** — 1 g of excess glue on an 8 g structure is 300 SE points.
3. **The tension-to-attach-block joint** — the most common cause of an early, inexplicable break.
4. **Correct bay spacing**, tuned by whether you are crushing- or buckling-limited.
5. **Jig accuracy and symmetry** — asymmetric structures fail at the stiffer side, well below capacity.
6. **The bonus decision**, made from test data rather than ambition.
7. Geometry micro-optimization — real, but the smallest lever of these seven.

Nationals-winning Division C boomilevers land in the **2,000–3,500 SE** range. The design above targets ~2,580 on paper. Reaching it takes five or more iterations, and the difference between a 1,200 structure and a 2,500 structure is almost never the drawing — it is wood selection, glue control, and joint craft.